

BioSpecInfo

Deploy Guide — BioSpecInfo

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Field	Value
Software	BioSpecInfo
Version described	bsi-v191
Purpose	Operational procedure for publishing, updating and rolling back the application.

0. What gets published, in practice

BioSpecInfo is a set of **static files**. There is nothing to compile, no process to keep running, no database to provision.

Build	None. The repository files <i>are</i> the application.
Server runtime	None
Environment variables	None for the application. Only the optional proxy has any (\$4).
Database	None server-side
Server requirement	Serve static files over HTTPS with the correct MIME types

The one non-obvious requirement: the server must return `application/wasm` for `.wasm` files, otherwise RDKit and SQLite do not start. GitHub Pages, Netlify, Vercel and Cloudflare Pages already do.

1. Environments

Environment	How it is obtained	What it is for
Development	<code>python3 -m http.server 8899</code> from the root	Daily work and verification benches
Preview	A branch published on Pages, or <code>npx serve</code>	Checking before merging
Production	GitHub Pages from the <code>main</code> branch	<code>samupropiol-ship-it.github.io/BioSpecInfo-v11/</code>

Why a server is needed even in development. Opening `index.html` with a double click (`file://`) does not work: RDKit and SQLite are WebAssembly modules and require an HTTP context. It is the reason the verification benches also run against a local server.

2. Publishing to production

2.1 Procedure

```
# 1. from the working branch, the full battery
python3 -m http.server 8899 &
node tools/genera-evidenza.js          # must close with 0 failures

# 2. bump the version – TWO lines, always as a pair
#   sw.js:      var CACHE = 'bsi-v<NNN+1>';
#   index.html: window.BSI_APP_VERSION='bsi-v<NNN+1>';

# 3. regenerate SBOM, PDFs, packages and evidence on the new version
node tools/genera-sbom.js
node tools/genera-pdf.js               # 36 PDFs; every footer carries version and commit
node tools/genera-pacchetti.js         # the three delivery packages; fails if a link is broken
node tools/genera-evidenza.js

# 4. merge into main
git add -A && git commit -m "..."
git push -u origin <branch>
#   then pull request and squash merge
```

GitHub Pages publishes automatically from `main`: the *pages build and deployment* workflow takes about a minute.

2.1-bis Publishing the version's release

The application is online as soon as `main` updates; the **release** serves a different purpose: giving an assessor a fixed point to download — a tag, the version notes and the delivery packages attached.

It is published from the repository, not from outside:

Actions → release → Run workflow, field `versione`: `bsi-v<NNN>`.

The workflow creates the tag on the current commit, opens the release and attaches the three packages from `consegna/`. The full archive — `models/` included, about 124 MB — is generated by GitHub itself as *Source code (zip)*.

Why from the repository and not from the command line. The credentials a remote session uses on the repository are authorised to write to branches and **not to tags**: the `POST` to `git-receive-pack` for `refs/tags/...` answers `403 Forbidden`, while a new branch goes through. The Actions token, by contrast, writes to the repository that hosts it.

What the workflow **refuses** to do:

Condition	Outcome
<code>CHANGELOG.md</code> has no <code>## [bsi-v<NNN>]</code> entry with content	fails: a release without notes looks like a version and does not say what changed

Condition	Outcome
fewer than three packages in <code>consegna/</code>	fails: a missing attachment would make the release silently incomplete
the tag does not point at the commit the packages came from	fails: a release must have a single provenance

That last row exists because of a real defect. On re-running the workflow to refresh the packages, `gh release edit` updated notes and assets but **did not move the tag**: the *Source code (zip)* archive, which GitHub generates from the tag, stayed at the first publication's commit while the packages came from a later one. Two provenances, and nothing saying so. The tag is now moved, and a following step does not take that on trust: it asks GitHub where the tag points and compares.

The workflow is **re-runnable**: if the release already exists it updates notes, assets and tag instead of failing halfway. Moving an already-published tag is not free — whoever downloaded it already has different content — so it applies to a release just published, not an old one: for that, publish a new version.

2.2 The two version lines

`CACHE` in `sw.js` and `BSI_APP_VERSION` in `index.html` **must match**.

- `CACHE` determines whether the Service Worker considers the cache stale:

without a bump, devices keep serving the old files and the publication reaches nobody.

- `BSI_APP_VERSION` is what the user sees in the "Updates" window and what the

update check compares with the `sw.js` on GitHub.

They drifted apart once in the past — the second had been stuck for months — and the result was that the update check was comparing the wrong number.

2.3 How the update reaches users

1. The browser detects a different `sw.js` and installs the new Service Worker.
2. `skipWaiting()` activates it at once; the old caches are deleted.
3. The Service Worker **notifies** the open tabs (`BSI_SW_UPDATED`) — it does not reload them on its own authority.

1. The page decides: if no work is in progress it reloads itself; if the assistant is answering or there is unsaved text, it shows "Update now / Later".

Whoever opens the app at that moment already receives the new version.

3. Publishing alternatives

The application does not depend on GitHub Pages. Any static file host will do, provided it is over HTTPS (the Service Worker requires it).

Platform	Command / configuration	Notes
GitHub Pages	Settings → Pages → branch <code>main</code> , folder <code>/</code>	In use. Free, custom domain possible
Netlify	<code>netlify deploy --prod --dir=.</code>	No build command
Vercel	<code>vercel --prod</code>	Static project, no framework
Cloudflare Pages	<code>npx wrangler pages deploy .</code>	Convenient if the proxy is already in use
VPS with nginx	Copy the files to <code>/var/www/biospecinfo</code>	See the configuration below

nginx configuration

```
server {
  listen 443 ssl http2;
  server_name biospecinfo.example.org;
  root /var/www/biospecinfo;

  # Without this MIME type, RDKit and SQLite do not start
  types { application/wasm wasm; }

  # sw.js must not be cached by the CDN: it is what announces new versions
  location = /sw.js {
    add_header Cache-Control "no-cache, no-store, must-revalidate";
  }

  location / {
    try_files $uri $uri/ /index.html;
  }
}
```

A rule that holds everywhere: `sw.js` must not be cached

If a CDN keeps `sw.js`, the browser keeps seeing the old version and **no update ever arrives**, however much one publishes. This holds on every platform.

4. Optional proxy — a separate deployment

The proxy is a **distinct** component, owned by the user, published on their own Cloudflare account. It is not required in order to use the application.

```
cd proxy
npx wrangler login
npx wrangler deploy
# the address is printed: https://spectra-proxy.<account>.workers.dev

npx wrangler secret put GROQ_KEYS      # one or more keys, comma-separated
npx wrangler secret put GEMINI_KEYS   # optional
```

The address is then pasted into the assistant's "Proxy" box, inside the app. The same four steps are available **inside the application**, with the commands ready to copy.

Recommended configuration

```
# Without this variable the proxy accepts requests from any origin
npx wrangler secret put ORIGINI
# value: https://samupropiol-ship-it.github.io
```

Variable	Default	Why change it
ORIGINI	(empty = all)	Always set it. An open proxy is usable by anyone, at the expense of your quota.
LIMITE_IP	20	Requests per IP within the window
TETTO_GIORNO	2000	Overall daily ceiling

Keys must never go into `wrangler.toml`. That file is versioned: a key written there ends up on GitHub, and the bots that scan it find it within hours. Use `wrangler secret put`, which keeps them out of the repository.

5. Domain and certificate

Aspect	Situation
HTTPS	Mandatory: without it the Service Worker does not register and the app does not work offline
Certificate	Provided and renewed automatically by GitHub Pages
Custom domain	Settings → Pages → Custom domain, plus a <code>CNAME</code> record pointing to <code><user>.github.io</code>

6. Monitoring

There is no server to monitor: there are no processes that could fall over, nor resources that could run out.

What to watch	How
Outcome of the publication	GitHub → Actions → <i>pages build and deployment</i>
Availability	An HTTP check on <code>index.html</code> (UptimeRobot or similar)
Client-side errors	The browser console. Remote telemetry exists but is disabled (<code>BSI_TELEMETRY_URL</code> empty): enabling it requires a deliberate decision about data handling.
Proxy consumption	Cloudflare → Workers → Analytics

7. Rollback and emergency procedures

7.1 Going back to the previous version

```
git revert <commit>
git push origin main
```

Pages republishes in about a minute. Devices receive the restored version as a normal update — **provided CACHE changes**: a revert that also takes the version number backwards is fine, because what counts is that the value be *different*, not that it be greater.

7.2 "I published but I do not see the changes"

In order:

1. **Was the version bumped?** It is the cause in ninety per cent of cases.

Without it, the Service Worker serves the cache.

1. **Has Pages finished?** GitHub → Actions.
2. **Is `sw.js` cached by a CDN?** It must answer `no-cache`.
3. From the app: ✨ menu → **Updates** → *Check*, which forces a `registration.update()`.

1. As a last resort: developer tools → Application → Service Workers →

Unregister, then reload.

7.3 "The assistant has stopped answering"

This is not a publishing problem: it concerns the AI provider. In the app, the 🦋 **Test** button measures in a few seconds which providers the device can actually reach, without needing any key.

7.4 Restoring the user's data

The data live on the device and **there are no copies on a server**: they cannot be restored remotely. This is a direct consequence of the local-first architecture, and it is declared to the user. Exports (notes, documents) remain the only form of backup.

8. Pre-publication checklist

- ☐ `node tools/genera-evidenza.js` closes with **0 failures**
- ☐ **CACHE** in `sw.js` and `BSI_APP_VERSION` in `index.html` bumped and **equal**
- ☐ `node tools/genera-sbom.js` re-run
- ☐ `node tools/genera-pdf.js` re-run — otherwise the attachments describe the previous version
- ☐ `node tools/genera-pacchetti.js` re-run and closed with **0 broken links**

- ☐ Evidence report regenerated against the commit being published
- ☐ Any new deviations noted in `09-Release-Conformance-Statement.md` §4
- ☐ No conflict markers (`verifica_guida` checks this across 58 files)
- ☐ No API key in the versioned files
- ☐ The accessibility debt **has not grown** — if it has fallen, regenerate the

baseline with `node tools/verifica-accessibilita.js --aggiorna-riferimento` and record it in the commit

- ☐ Code coverage **has not fallen** — `audit_copertura` fails if it has

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